THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** glow dimly	4.5 V is less than the rated 6 V.
2	**(b)** test with a tester; add salt to water to make it conduct	Dissolved salts supply ions.
3	**(c)** a conductor	Current passes through it.
4	**(c)** glow very bright and may fuse	Exceeds rating.
5	**(a)** tap water has dissolved salts	Dissolved ions conduct.
6	**(b)** an insulator	Current does not pass.
7	**(c)** series	Single path.
8	**(a)** copper conducts better	Brightness suggests current.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Current flows only in a complete (closed) circuit, so R explains why the bulb glows.
10	**(c)** A is true but R is false.	A fuse is in series, so A is true; but fuse wire has a LOW melting point and melts when current is too large, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tungsten has a very high melting point (1); Copper would melt before glowing (1).	1+1
12	Metals are good conductors (1); They carry current to the appliance (1).	1+1
13	Prevent touching live parts (1); Avoid shocks (1).	1+1
14	Nichrome gets very hot when current passes (1); It does not melt or oxidise easily at high temperatures (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	6 x 1.5 = 9 V (1); Much higher than 3 V (1); Bulb may fuse (1).	1+1+1
16	They need a large current (1); Thin wiring may overheat (1); Risk of fire (1).	1+1+1
17	Switch off the supply (1); Dry hands (1); Use proper bulb (1).	1+1+1
	OR: Thin wire heats more (1); Melts first (1); Protects the circuit (1).	1+1+1
18	Iron would melt or burn (1); Tungsten has a high melting point (1); Bulb has inert gas (1).	1+1+1
	OR: One path (1); Same current (1); If one fails all stop (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $1.5 + 1.5 = 3$ V (1). (ii) The switch (1). (iii) The filament (1). (iv) The filament may be broken (fused) or there may be a loose connection (1); in both cases the circuit is not complete, so no current flows through the filament (1).	1+1+1+2
20	Switch off unused devices (1); Use LED bulbs (1); Use daylight (1); Do not overload (1); Service appliances (1).	1+1+1+1+1
	OR: Overloading (1); Short circuits (1); Damaged wires (1); Use ISI products (1); Use fuse or MCB (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the nail is coated with paint or rust	Surface layer blocks contact.
2	**(d)** goes off	The loop is broken.
3	**(c)** its melting point is lower than that of the connecting wires	Low melting.
4	**(c)** touch it	Touching is unsafe.
5	**(b)** the reversed cell reduces the total voltage	Opposed cells cancel partly.
6	**(d)** the lemon juice conducts	Ions in the juice.
7	**(b)** to avoid contact and shock	Safety.
8	**(c)** a broken connecting wire	Open path.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the plastic covering is used because it is an insulator, which R does not explain.
10	**(d)** A is false but R is true.	A gap makes an open circuit with no current, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cells in series add their voltages (1); Higher voltage drives more current through the filament (1).	1+1
12	Plastic is an insulator (1); It protects the user from shocks (1).	1+1
13	15 A fuse will not protect thin wires (1); They may overheat (1).	1+1
14	It can cut off the current to the appliance (1); It controls the appliance (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	9 / 1.5 = 6 (1); Six cells in series (1); + to - connections (1).	1+1+1
16	Switch off appliances (1); Replace the fuse with the same rating (1); Avoid overloading (1).	1+1+1
17	Chemicals may leak (1); Damage to the toy (1); Cell is saved (1).	1+1+1
	OR: Opposing cells reduce voltage (1); Little or no current (1); Filament does not heat (1).	1+1+1
18	Use less energy (1); Less fuel burnt (1); Fewer emissions (1).	1+1+1
	OR: Saves electrical energy (1); Reduces bills (1); Saves resources (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The fuse (1). (ii) It has a low melting point (1). (iii) The heating effect of electric current (1). (iv) Copper has a much higher melting point, so it would not melt when the current is too large (1); the wires would stay overheated and could cause a fire (1).	1+1+1+2
20	Insulated wiring (1); Fuse and MCB (1); Switches in good condition (1); Dry hands (1); Switch off and call help (1).	1+1+1+1+1
	OR: Light at night (1); Machines run (1); Communication (1); Rules prevent accidents (1); Careful use (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** iron nail, rubber, aluminium foil	Conductor, insulator, conductor.
2	**(a)** heating effect	Wire melts by heat.
3	**(d)** a sealed metal sheath with insulating powder	Safe design.
4	**(c)** it can carry more current without overheating	Thick wires heat less.
5	**(c)** a long wire fits in a small space	Compact design.
6	**(a)** switching to LED bulbs	Efficient.
7	**(d)** water conducts and can give a shock	Safety.
8	**(a)** connecting many appliances to one socket	Large current.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Removing a bulb breaks the only path, so no current flows anywhere in the circuit.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but reusability does not explain why it switches off.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Current flows quickly through the circuit (1); All parts conduct (1).	1+1
12	Positive of one to negative of next (1); 3 x 1.5 V = 4.5 V (1).	1+1
13	Thin wires overheat (1); Safe current carrying (1).	1+1
14	Conduct and heat (1); Can be shaped as coils (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Large current (1); Wires heat (1); Insulation burns (1).	1+1+1
16	Distilled water has no dissolved salts (1); Tap water has dissolved salts (1); So it conducts (1).	1+1+1
17	Lighter (1); Cheaper (1); Good conductor (1).	1+1+1
	OR: Dip tester ends (1); Bulb glows for conducting liquids (1); Lemon and salt solution conduct, sugar does not (1).	1+1+1
18	More current more heat (1); Heat increases with current (1); Fuse chosen accordingly (1).	1+1+1
	OR: Overheating (1); Damage to holder and wiring (1); Fire (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Series connection (1). (ii) There is only one path for the current, and a fused bulb breaks it (1). (iii) The other bulbs would keep glowing (1). (iv) Each appliance can be switched on or off independently (1), and each gets the full voltage of the supply (1).	1+1+1+2
20	Tester shows conduction (1); Metals light the bulb (1); Poor conductors give a weak current (1); Needs good contacts (1); Cannot measure resistance (1).	1+1+1+1+1
	OR: In series with the appliance (1); Not across the appliance (1); Current passes through it (1); Melts on overload (1); Appliance isolated (1).	1+1+1+1+1

Mistakes to avoid:

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- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

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